

Prompt Injection Lab — NICE Framework Alignment

Teacher Implementation Guide — Rita Cyber Ed — Miss Sabri

Connecting K-12 cybersecurity instruction to the official NICE Workforce Framework for Cybersecurity

What is this lab? Students act as AI red team specialists, attempting to manipulate a bank customer service chatbot (“Ava”) into breaking its hidden security rules. This teaches prompt injection, system prompt leakage, and sensitive information disclosure — OWASP’s top LLM vulnerabilities (2025) — through adversarial hands-on practice.

Technical requirements: A browser. That’s it. No API key. No software install. No login. Runs entirely client-side. Works on Chromebooks, school iPads, and any DCPS-issued device.

Live demo: <https://ritasabri.github.io/rita-cyber-ed/>

1. Where YOU Fit: NICE Work Roles for Educators

Before mapping student outcomes, it’s worth noting that the NICE Framework explicitly recognizes the work *you* do as a cybersecurity educator. These are official federal workforce categories:

Work Role ID	Role	What You Do
OG-WRL-004	Cybersecurity Curriculum Development	Develop, plan, coordinate, and evaluate cybersecurity awareness, training, and education content. <i>This lab is curriculum development work.</i>
OG-WRL-005	Cybersecurity Instruction	Develop and conduct cybersecurity awareness, training, and education. <i>Teaching this lab is direct instruction work.</i>

Position yourself this way when speaking to your administrators: you are filling a documented federal workforce category. The NICE Framework is the official taxonomy used by federal cybersecurity hiring.

2. Where STUDENTS Fit: NICE Work Roles This Lab Introduces

This lab gives students a hands-on introduction to specific NICE Work Roles in the *Protection & Defense (PD)* and *Design & Development (DD)* categories:

Work Role ID	Role	How This Lab Connects
PD-WRL-001	Defensive Cybersecurity	Students analyze attack data and identify mitigation strategies (Section B Q4 of student handout)
PD-WRL-003	Incident Response	Students experience the attacker’s perspective so they can later respond to real incidents
PD-WRL-006	Threat Analysis	Students analyze WHY attacks succeed — the core skill of threat analysts (Section B Q2)
PD-WRL-007	Vulnerability Analysis	Students systematically test for weaknesses — the core skill of vulnerability analysts (Section A Attack Log)

Work Role ID	Role	How This Lab Connects
DD-WRL-003	Secure Software Development	Students propose mitigations a developer would implement (Section B Q4)
DD-WRL-005	Software Security Assessment	Students perform adversarial testing of a software product — exactly what this role does (Section C Creative Extension)
DD-WRL-007	Systems Testing & Evaluation	Students plan, execute, and report on system tests (entire lab structure)

Emerging Role: AI Red Team Specialist. While not yet a formal NICE Work Role, SANS Institute's 2026 Workforce Report found that **23% of cybersecurity organizations are actively hiring** for this position. The skills this lab teaches — adversarial testing of AI systems — map directly to this emerging role. Students who complete this lab are practicing skills the industry is hiring for right now.

3. NICE TKS Statements Demonstrated in This Lab

These are the official NICE Framework Task, Knowledge, and Skill (TKS) statements your students will demonstrate competency in:

Code	Type	Statement
K0005	Knowledge	Knowledge of cyber threats and vulnerabilities
K0070	Knowledge	Knowledge of system and application security threats and vulnerabilities
K0106	Knowledge	Knowledge of what constitutes a network attack and its relationship to threats and vulnerabilities
K0119	Knowledge	Knowledge of hacking methodologies
S0025	Skill	Skill in detecting host and network-based intrusions via intrusion detection technologies
S0078	Skill	Skill in recognizing and categorizing types of vulnerabilities and associated attacks
T0175	Task	Perform real-time cyber defense incident handling tasks
T0260	Task	Analyze identified malicious activity to determine weaknesses exploited, exploitation methods, and effects on system

4. Lab Activity → TKS & Work Role Crosswalk

Student Activity	NICE TKS	Work Role	OWASP
Document attack attempts in structured log (Section A)	T0260	PD-WRL-007	—
Identify which attack succeeded and why (Section B Q1-2)	S0078, T0260	PD-WRL-006	LLM01
Categorize attack using OWASP taxonomy (Section B Q3)	K0070, S0078	PD-WRL-007	LLM01/02/07
Propose specific remediation (Section B Q4)	T0175, K0070	DD-WRL-003	—
Design original creative attack (Section C)	K0119, S0078	DD-WRL-005	LLM01
Analyze real-world implications (Section D Q7)	K0005, K0070	PD-WRL-003	—
Connect to career pathways (Section D Q8-9)	K0005	OG-WRL-005	—

5. OWASP Top 10 for LLM Applications (2025) Coverage

LLM01 — Prompt Injection *(Primary Focus)*

Manipulating an AI system to ignore, override, or bypass its instructions. Students practice this directly through role-play, authority claims, instruction overrides, and creative framing attacks.

LLM02 — Sensitive Information Disclosure *(Secondary Focus)*

Getting an AI to reveal confidential data it was instructed to protect. Students attempt to extract the hidden employee access code (VAULT-7829-ALPHA).

LLM07 — System Prompt Leakage *(Secondary Focus)*

Getting an AI to reveal its own hidden instructions. Students use indirect questioning, summary requests, and rule-listing techniques to expose the system prompt.

6. Suggested Lesson Flow (45 Minutes)

Time	Phase	What Happens
0–5 min	Framing	Define prompt injection. Show OWASP Top 10 for LLMs. Hook: “23% of orgs are hiring AI red team specialists right now.”
5–10 min	Mission Brief	Open the lab. Students read the mission. Set expectations: document everything.
10–25 min	Red Team Phase (Sections A–B)	Students work individually or in pairs. They discover attack techniques themselves. The handout does NOT give them attack examples.
25–35 min	Creative Phase (Section C)	Students pick a category and design an ORIGINAL attack, then test it. This is where the strongest learning happens.
35–40 min	Real-World Reflection (Section D)	Students transfer learning to the hospital scenario.
40–45 min	Debrief	2–3 students share their best attack. Discuss careers introduced in the “Did You Know” boxes.

7. Differentiation Options

Student Group	Modification
Struggling learners / younger students	Toggle Easy Mode in lab settings. The bot will crack on almost any attempt, allowing students to focus on the analysis phase.
English Language Learners (MLLs)	Pair with bilingual peer. Allow analysis questions in primary language with English vocabulary key terms.
Advanced students / honors classes	Disable Easy Mode AND disable Hints. Require all 7 attack categories to be discovered (visible in scoreboard).
Students with IEPs	Provide the handout digitally. Allow voice-to-text. Reduce Section A requirement from 5 attempts to 3.

8. Assessment Rubric (20 Points)

Criterion	5 pts — Exemplary	3 pts — Proficient	1 pt — Developing
Persistence (Section A)	5+ documented attempts using varied techniques	3–4 attempts with some variety	1–2 attempts, single technique
Analysis (Section B)	Identifies specific mechanism of attack and proposes realistic fix	General understanding of why attack worked	Surface-level or missing analysis
Creative Attack (Section C)	Original attack distinct from earlier attempts; works OR thoughtful reflection on why not	Attempted creative attack with some originality	Copy of earlier attempt OR missing
Real-World Transfer (Section D)	2+ specific risks identified; thoughtful career reflection	General risks identified	Vague or missing

9. Free Resources

- **Prompt Injection Lab**
ritasabri.github.io/rita-cyber-ed
- **NICE Framework Resource Center**
nist.gov/itl/applied-cybersecurity/nice
- **NICE Work Roles Browser**
niccs.cisa.gov/tools/nice-framework
- **OWASP Top 10 for LLM Applications**
genai.owasp.org

- **CyberSeek Career Pathways**
cyberseek.org
- **SANS 2026 Workforce Report**
sans.org/white-papers

- **Google Cybersecurity Forecast 2026**
cloud.google.com/security
- **KC7 Threat Investigation Game**
kc7cyber.com

About the Curriculum Patch: This lab is part of the Curriculum Patch framework — a methodology for recontextualizing existing CS curricula to align with the NICE Cybersecurity Workforce Framework and modern AI threats. You don't need to build new courses — you patch what you have.

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